



AI Challenges in Mathematical Investing

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Seminar's Objective

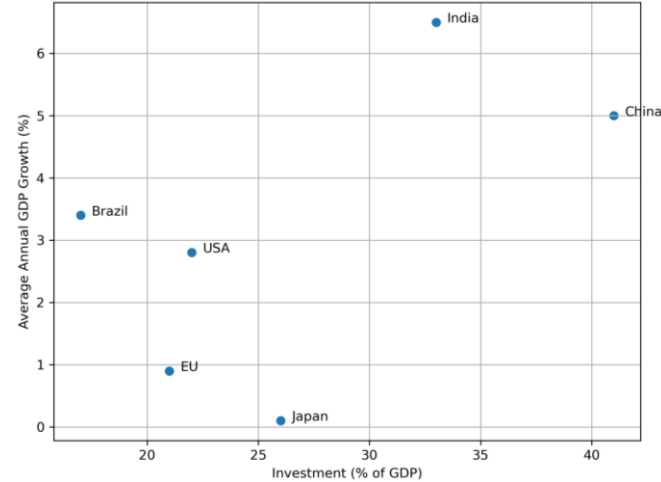
- Investing can be viewed as a **complex data science problem**
- General-purpose AI approaches often fall short in practice
- This talk explores the key reasons why various mathematical and AI-driven methods fail in the context of investing
- **Objective: Offer insights into how these challenges can be mitigated through agentic solutions**



Billions of people can drive a car, however only a small number of investors consistently beat the collective wisdom of the markets. This highlights the complexity of this task, and the difficulties we face when developing investment models.

Why Investing Matters

- **Investing** is the allocation of scarce resources today with the *expectation* of future benefits
 - Allocation decisions are made subject to high uncertainty
- **Efficient investing** is essential to a prosperous, self-reliant and sustainable society
 - Supports national security and sovereignty
 - Delivers stable job creation and social harmony
 - Is the foundation for economic competitiveness and innovation leadership
- In an increasingly multi-polar world, efficient investing is a competitive advantage



Source: <https://data.worldbank.org/>

Higher investments support stronger GDP growth. However, as the plot shows, not all investments are equally efficient in terms of generating growth.

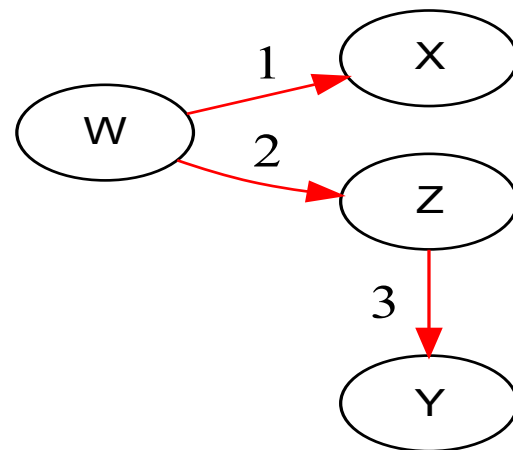
The discipline of investing can help Governments and private institutions achieve superior outcomes.

Investing has become a Data Science Problem

Conditions	Investing is largely driven by speculation, due to - no financial theories - no computers - simplistic math models - limited data	Academics formalize the concepts of diversification, risk-premia and valuation MPT, CAPM, APT, risk factors, Black-Scholes, market microstructure	Technological advances in - data storage - supercomputing - networking Math models become more sophisticated	The explosion in alternative data shifts research objectives, from valuation and forecasting, towards nowcasting (direct estimation)
	1950 2000 2015 Technical analysis Fundamental analysis, econometrics Market microstructure, HFT Machine learning, causal inference			
Rationale	Investment decisions are motivated by stories or themes	Investment decisions rely on financial analysis of specific opportunities	Investment decisions increasingly rely on research and technology. Some of the best performing funds are quant-driven	
Training	Investment professionals do not receive a formal education	During this period, CFA is the gold standard of financial accreditation	Firms hire STEM grads for research and software engineering roles	High demand for data scientists and automation experts

However, Accurate Forecasting is Not Enough

- Making investment decisions requires a causal attribution of premia to risk factors
 - Without this attribution, investors are exposed to unrewarded risks
- Investing is not a forecasting problem
 - Accurate forecasts are not enough in investing
- This causal attribution must be done under uncertainty
 - What are the premia? (expected returns)
 - What are the risk factors? (risk model)
 - What risk causes a particular premium? (attribution)
 - What are the interactions across risk factors (joint effects)



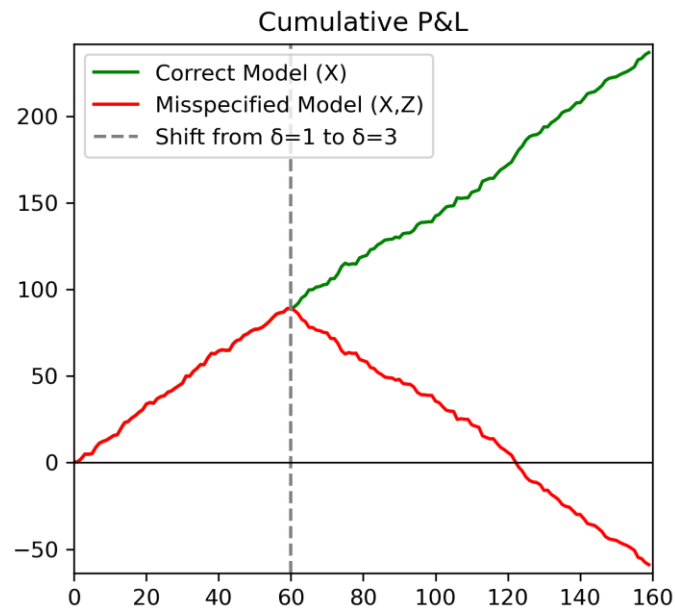
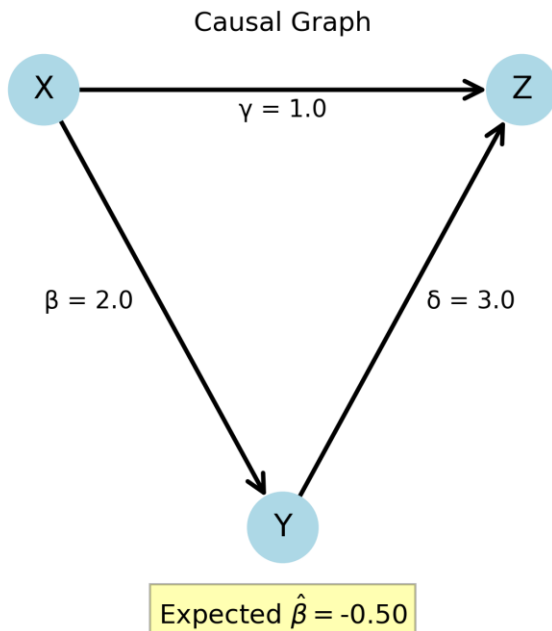
Two variables can be associated without one causing the other. For example, weather (W) influences ice cream sales (X) and the number of swimmers (Z), hence the number of drownings (Y).

An AI algorithm can use X to predict Y , however investing requires estimating the effect of X on Y .

Example: Causal Factor Investing

The causal graph shows that: (a) factor X (the cause) influences variable Y (the target) with factor loading $\beta = 2$ (the effect), and (b) that variable Z is influenced by variables X and Y , thus Z is a collider.

Applying [causal discovery tools](#), a researcher will exclude Z as a control. The green line shows that the correctly specified factor strategy is equally profitable regardless of the effect size of Y on Z ($\delta = 1$ or $\delta = 3$).



When $\delta = 1$, then $E[\hat{\beta}|X, Z] = 0.5$, and the misspecified strategy still performs well (see red line before the dashed vertical line), because the estimated $\hat{\beta}$ has the correct sign. However, if the value of δ changes from $\delta = 1$ to $\delta = 3$, then $E[\hat{\beta}|X, Z] = -0.5$, and [the misspecified strategy yields systematic losses](#) (see red line after the dashed vertical line), because an investor seeking exposure to X will sell securities that should be bought and vice versa.

Problem #1: Barriers to Experimentation

- Scientists propose causal theories to explain phenomena (observed associations)
 - A theory that cannot be falsified empirically, is not scientific
 - Note: Not scientific does not mean false
- **Association does not imply causation**
 - X is associated with Y iff $P[Y, X] \neq P[X]P[Y]$
 - X causes Y iff $P[Y|do[X]] > P[Y]$
- In the natural sciences, causal theories are falsified through randomized controlled experiments
- Controlled financial experiments are rarely possible
 - E.g., we cannot repeat the Flash Crash of 2010 controlling for some market participants
 - **Instead, researchers engage in uncontrolled multiple testing**



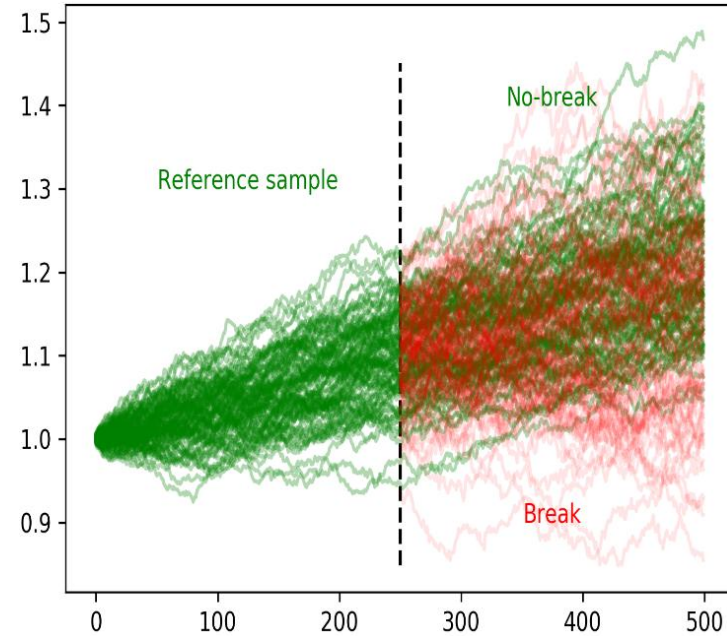
In the 1930s, Ronald Fisher popularized randomized experiments as a way to de-confound variables.

The first published RCT appeared in 1948. Today, well-blinded RCTs are considered the gold standard in experimental research.

Instead of RCTs, financial researchers often compute backtests (historical *simulations*).

Problem #2: Non-Stationarity

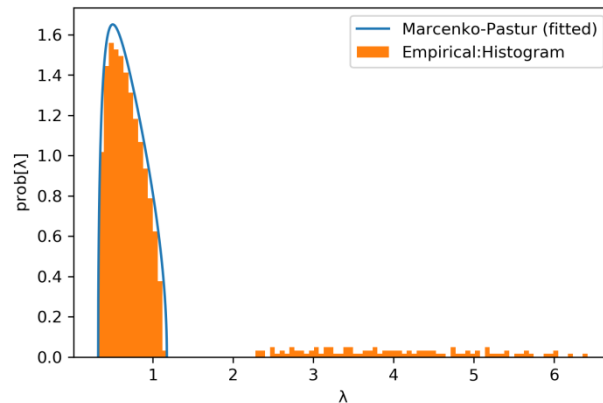
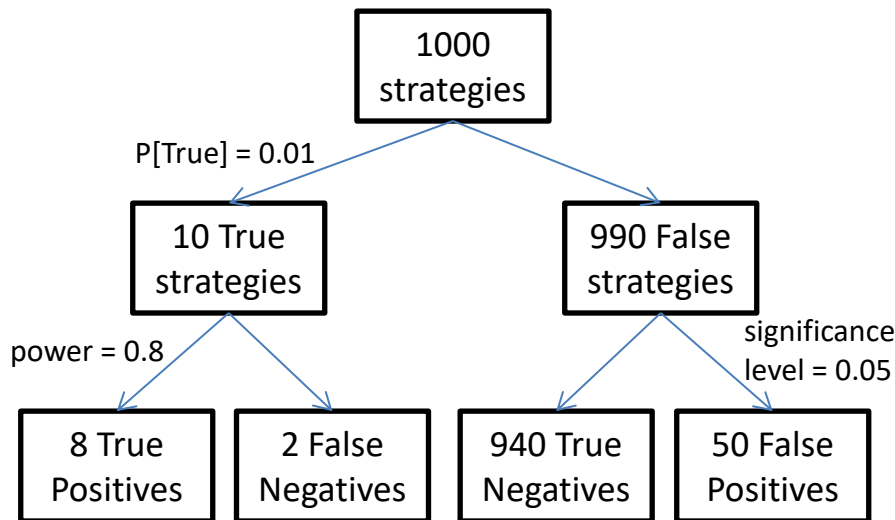
- Time series analysis relies on two key assumptions
 - **Ergodicity**: Observations span the entire support for the stochastic process
 - **Stationarity**: The unconditional joint probability distribution is time invariant
- Ergodicity can be recovered by combining non-ergodic processes
 - E.g., investing in diversified indices, rather than in an individual company that may go bankrupt
- Financial systems are non-stationary due to
 - **structural breaks** (e.g., changes in regulations)
 - **drift in parameters** that characterize the data-generating process (e.g., competition)



After 250 observations, a structural break takes place with probability 0.5. The problem is to detect the break as soon as possible, with high precision and high recall. See [ADIA Lab Data Science Tournament](#).

Problem #3: Low Signal-to-Noise Ratio

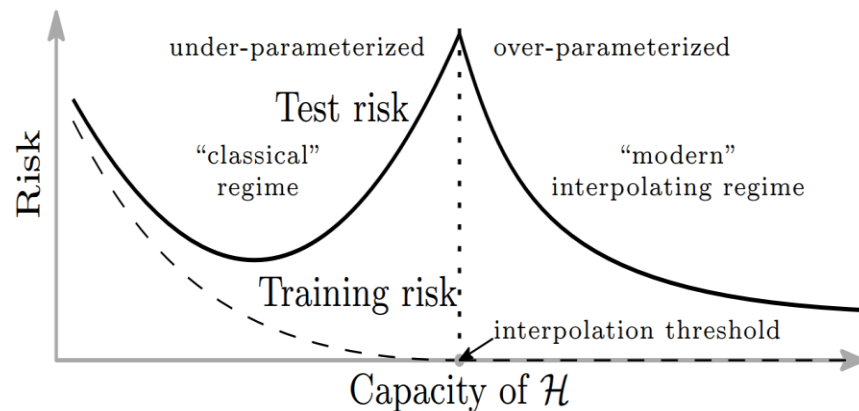
Strict competition among investment managers lowers the signal-to-noise ratio. For example, almost all the information contained in a factor covariance matrix (e.g., Barra) can be associated with noise rather than signal. As a result, the probability of finding a profitable investment strategy is extremely low, and alpha decays quickly.



Suppose that the probability of a backtested strategy being profitable is 1%. Then, at the standard thresholds of 5% significance and (optimistic) 80% power, researchers are expected to make 58 discoveries out of 1000 trials, where 8 are true positives and 50 are false positives. Under these circumstances, **a p-value of 5% implies that at least 86% of the discoveries are false!**

Problem #4: Small Samples

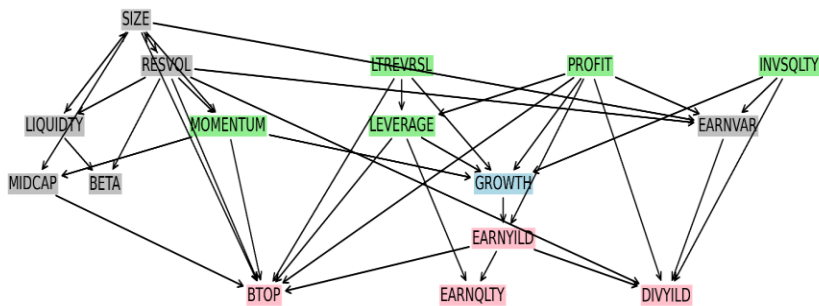
- Financial datasets often exhibit
 - Short time series, few variables
 - Highly cross-dependent variables
 - Highly serially-dependent variables
- Financial researchers often must produce inference from small samples
 - Strong train-set overfitting
 - Strong test-set overfitting
- At standard confidence levels, statistical tests on financial datasets have low power
 - High proportion of false negatives
 - High false discovery rate



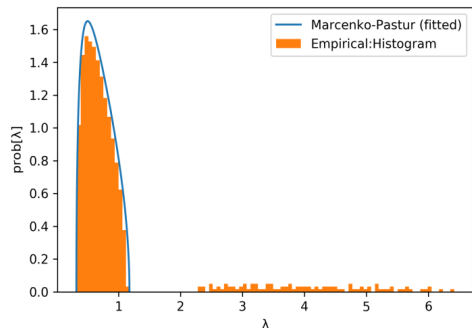
[Belkin et al. \[2019\]](#) propose the *possibility* of a “double descent” by fitting a model with thousands (possibly millions) of random Fourier features (RFF), in an attempt to achieve a lower mean squared error out-of-sample. Even if successful, the resulting model is a black-box: uninterpretable, and vulnerable to regime shifts, structural breaks and black-swans.

While massive overfitting may have commercial applications (e.g., marketing), it has limited value for institutional investors, for scientific and legal reasons (e.g., fiduciary duty).

Investing Requires a New Type of AI



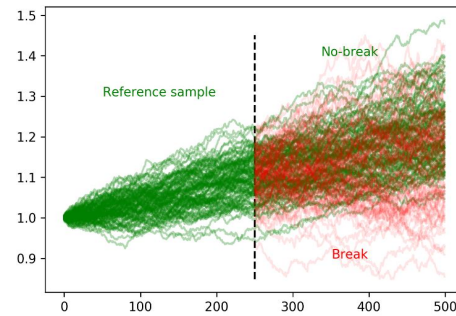
Investing is a causal problem



Low signal-to-noise ratio,
due to intense competition



Barriers to experimentation



Non-stationarity, due to
structural breaks and
parameter shift

$$N \gg T$$

The number of variables (N) greatly exceeds the number of
observations per variable (T)

The Use and Misuse of AI in Investing

Three questions to ask when presented with an AI investment solution:

1. Have you controlled for backtest overfitting?

- Train-set overfitting: Sensitivity analysis
- Test-set overfitting: Multiple-testing adjustment ([Deflated Sharpe Ratio](#))
- Out-of-sample evidence: Embargo

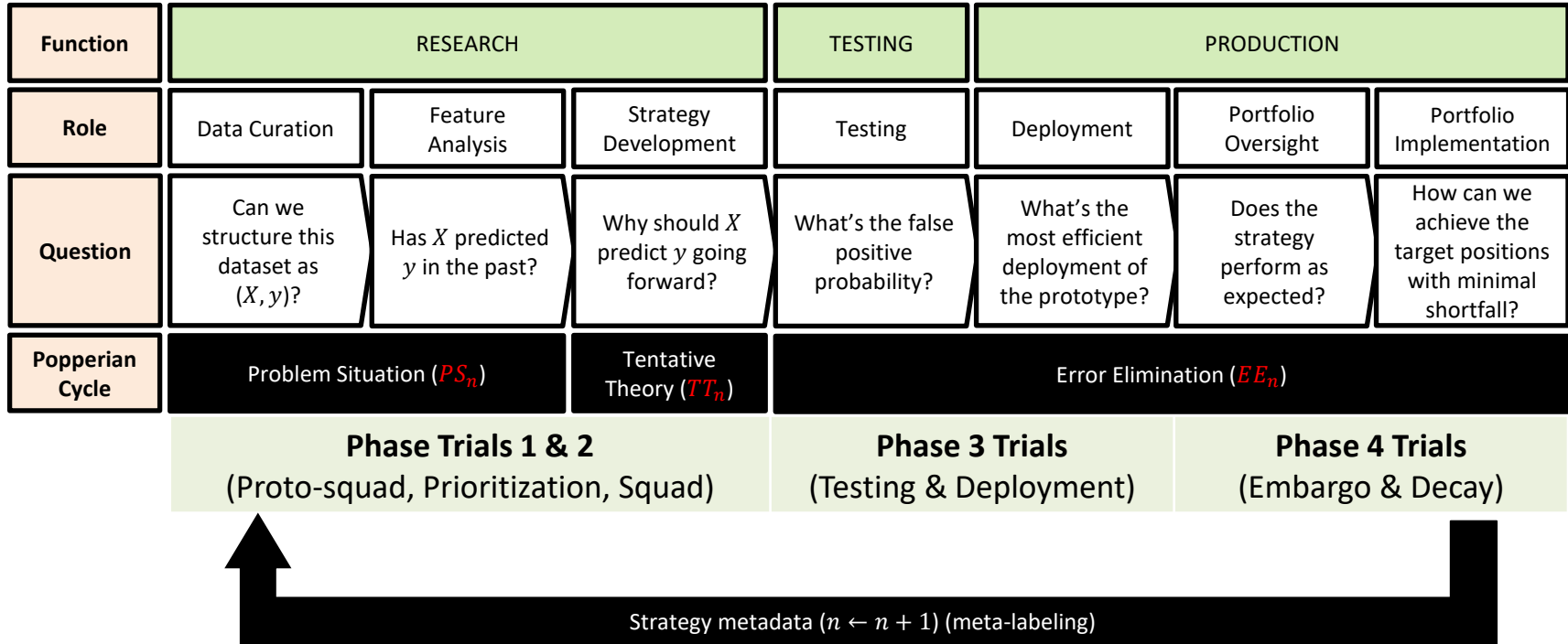
2. What is the causal mechanism responsible for this anomaly?

- Risk premia attribution: Causality is a necessary condition for investment efficiency
- Counterfactual reasoning:
 - prepare for black-swan events
 - coherent risk analysis & stress testing

3. Is this an ensemble method?

- Avoid [single point of failure](#)
- Bagging, boosting, stacking, [meta-labeling](#)

Structure of an Investment Assembly Line



For a description of the assembly line stations, read https://ssrn.com/abstract_id=3104847

The Assembly Line In Action

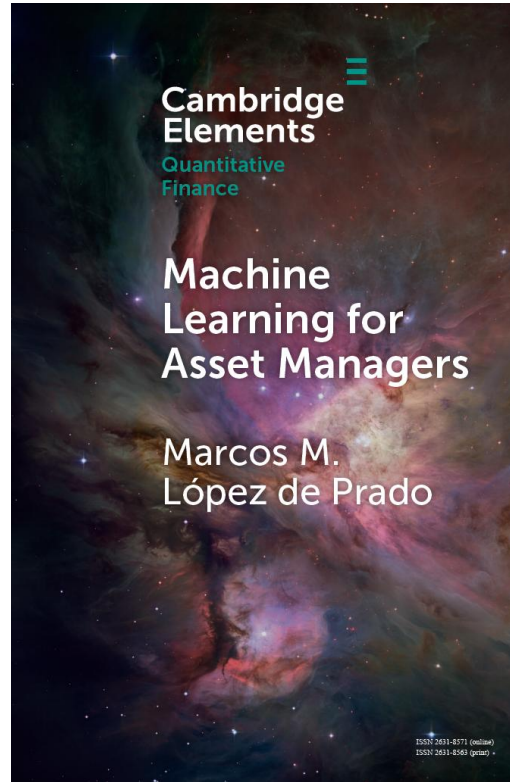
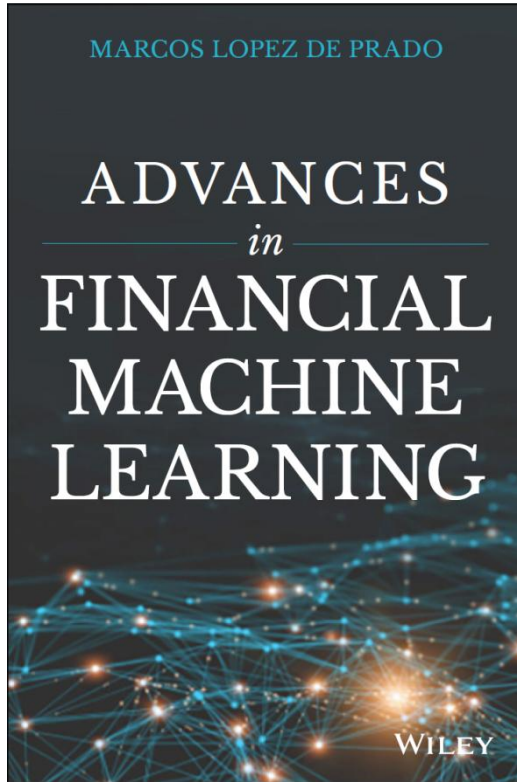
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For Additional Details



The first wave of quantitative innovation in finance was led by Markowitz optimization. Machine Learning is the second wave and it will touch every aspect of finance. López de Prado's Advances in Financial Machine Learning is essential for readers who want to be ahead of the technology rather than being replaced by it.
— Prof. **Campbell Harvey**, Duke University. Former President of the American Finance Association.

Financial problems require very distinct machine learning solutions. Dr. López de Prado's book is the first one to characterize what makes standard machine learning tools fail when applied to the field of finance, and the first one to provide practical solutions to unique challenges faced by asset managers. Everyone who wants to understand the future of finance should read this book.

— Prof. **Frank Fabozzi**, Johns Hopkins University. Editor of The Journal of Portfolio Management.

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